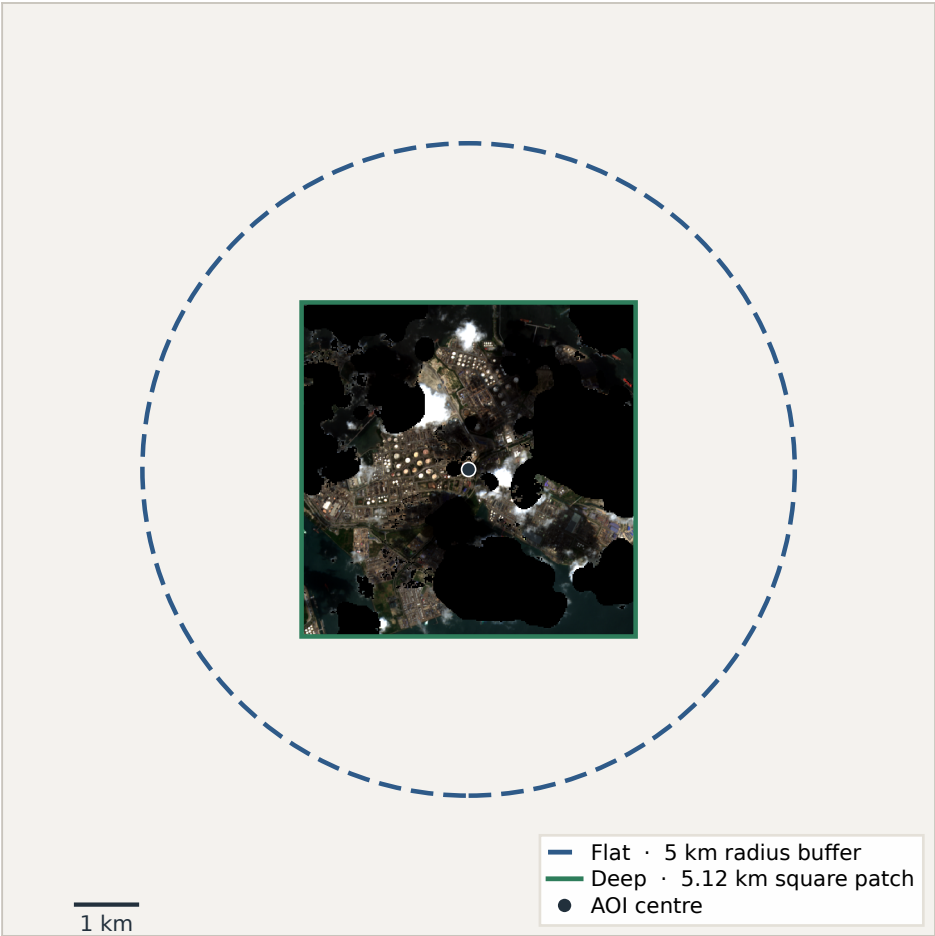


Figure 3.4 Worked example · Jurong Island (P004), forecast origin 14 March 2025

Latest eligible observations under the source-specific availability lags.

(a) Spatial supports · January 2025 Sentinel-2 composite

Under the locked-lag rule, the February 2025 composite was not yet eligible at this forecast origin. January inputs were eligible from 15 February and carried across the four-week input window.



(b) Remote-sensing representations

Flat

Sentinel-2 + VIIRS scalar anomalies

5 km buffer · five site-level scalars

NDVI anomaly	+0.56
NDWI anomaly	-0.40
NDBI anomaly	-0.93
BSI anomaly	-0.76
NTL anomaly	+0.36

Deep

Sentinel-2 representation

Sentinel-2 patch → resize to 224 × 224

→ frozen Prithvi → mean pooling

→ 1024-d embedding

224 × 224

(c) Shipping graph inputs around Jurong

GFW voyage edges: week ending 28 February
PortWatch tanker calls: week ending 7 March.

Dynamic GFW voyage edges

16 directed lanes · 257 voyages

busiest four shown

Origin → destination	Voyages
Ningbo → Jurong	58
Jurong → Ningbo	45
Fujairah → Jurong	26
Jurong → Fujairah	23

Fixed corridor edge

Jurong Malacca

Selected node attribute

Jurong

PortWatch tanker calls · 520